

Milestone 3: Deep Learning Model & Explainability

Objective

The objective of this milestone is to design and evaluate a **deep learning–based approach** for identifying the source scanner device used to scan a document. Unlike traditional machine learning models that rely on handcrafted features, this milestone leverages a **Convolutional Neural Network (CNN)** to automatically learn scanner-specific artifacts directly from raw scanned images. Additionally, **explainability techniques (Grad-CAM)** are applied to interpret the model's predictions.

Dataset Used

- **Type of images:** Official document scans
- **Scanners considered:**
 - HP
 - Canon 9000
 - Epson V550
- **Image format:** Converted to PNG
- **Image size:** 224 × 224
- **Total samples:** Balanced across all three scanners

The dataset was preprocessed in **Milestone 1** and reused for CNN training in this milestone.

Data Augmentation:

To improve generalization and reduce overfitting, the following augmentation techniques were applied during training:

- Random rotation

These augmentations help the model become robust to minor variations introduced during scanning.

CNN Architecture

A custom **Convolutional Neural Network (CNN)** was designed with the following structure:

- Input layer: 224×224×3
- Three convolutional blocks:
 - Conv2D → Batch Normalization → MaxPooling
- Fully connected layers:
 - Dense (128 units) with ReLU activation

- Dropout (0.5) to reduce overfitting
- Output layer:
 - Dense layer with Softmax activation (3 classes)

Model Summary:

- **Total parameters:** ~11 million
- **Optimizer:** Adam
- **Loss function:** Categorical Crossentropy

The explicit input layer ensured proper graph construction, which was essential for explainability using Grad-CAM.

Model Training

- **Train–Validation split:** 80% training, 20% validation
- **Batch size:** 32
- **Epochs:** 20–30
- **Hardware:** Google Colab (GPU)

Training performance was monitored using:

- Accuracy
 - Loss curves (training vs validation)
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Model Evaluation

Quantitative Metrics:

- **Training Accuracy:** ~97%
- **Validation Accuracy:** Above 95% (peak)
- **Metrics evaluated:**
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - Confusion Matrix

The CNN significantly outperformed baseline models by learning scanner-specific noise patterns directly from images.

--- Milestone 3: Final Classification Report ---				
	precision	recall	f1-score	support
HP	0.99	1.00	0.99	640
Canon9000-1	0.99	0.97	0.98	640
EpsonV550	0.97	0.98	0.98	640
accuracy			0.98	1920
macro avg	0.98	0.98	0.98	1920
weighted avg	0.98	0.98	0.98	1920

Explainability Using Grad-CAM

To interpret the CNN’s predictions, **Grad-CAM (Gradient-weighted Class Activation Mapping)** was applied.

Method:

- The final convolutional layer was automatically identified.
- Grad-CAM heatmaps were generated for validation images.
- Heatmaps highlight regions of the image that contributed most to the prediction.

Observations:

- The model focuses on subtle texture and noise patterns rather than document content.
- Scanner-specific artifacts are consistently highlighted across samples.

This confirms that the model learns **scanner signatures**, not semantic document features.

Results & Key Insights

- CNN successfully learned scanner-specific characteristics.
- Achieved **97% accuracy**, satisfying evaluation requirements.
- Grad-CAM provided transparency and validated model behavior.
- Data augmentation improved robustness but excessive augmentation reduced validation stability, requiring careful tuning.

Conclusion

Milestone 3 successfully demonstrates the effectiveness of deep learning for forensic scanner identification. The CNN-based approach achieves high accuracy and, with the integration of explainability techniques, ensures that predictions are interpretable and reliable. This milestone forms a strong foundation for deployment and real-world forensic applications.